

January 2026

IPC Online

Section: C1

Time: 30 min

Run four three threads.

- Thread 1 prints letter "A" one letter at a time
- Thread 2 prints letter "B" one letter at a time
- Thread 3 prints letter "C" one letter at a time
- Thread 4 prints a newline when a certain condition is met

Each line contains three "A"s, four "B"s, and five "C"s. The letters can appear at any order. Thread 4 prints a newline when the total count of letters in a line equals twelve. There will be seven lines printed in the output.

Implement using semaphores only , avoid busy waiting. For the count of A, B, Cs and the number of iterations, you must use counting semaphores. [Hint: You will need four counting semaphores.]

Sample Output:

AAABBBBCCCCC

ABCBABCBCACC

.....

BACABCBACBCC

Create a file <student_id>.cpp and submit it.